

Question 6. User Orders Count in 2019 | Left Join with Filtering and Aggregation (Medium)
(AMAZON,GOOGLE,UBER)

(REF :

<https://leetcode.com/explore/featured/card/sql-language/684/sql-relationship/4340/>)

Table: Users

+-----+		
Column Name	Type	
+-----+		
user_id	int	
join_date	date	
favorite_brand	varchar	
+-----+		

user_id is the primary key (column with unique values) of this table.

This table has the info of the users of an online shopping website where users can sell and buy items.

Table: Orders

+-----+		
Column Name	Type	
+-----+		
order_id	int	
order_date	date	
item_id	int	
buyer_id	int	
seller_id	int	
+-----+		

order_id is the primary key (column with unique values) of this table.

item_id is a foreign key (reference column) to the Items table.

buyer_id and seller_id are foreign keys to the Users table.

Table: Items

+-----+		
Column Name	Type	
+-----+		
item_id	int	
Item_brand	varchar	
+-----+		

item_id is the primary key (column with unique values) of this table.

Write a solution to find for each user, the join date and the number of orders they made as a buyer in 2019.

Return the result table in any order.

The result format is in the following example:

Example 1:

Input:

Users table:

user_id	join_date	favorite_brand
1	2018-01-01	Lenovo
2	2018-02-09	Samsung
3	2018-01-19	LG
4	2018-05-21	HP

Orders table:

order_id	order_date	item_id	buyer_id	seller_id
1	2019-08-01	4	1	2
2	2018-08-02	2	1	3
3	2019-08-03	3	2	3
4	2018-08-04	1	4	2
5	2018-08-04	1	3	4
6	2019-08-05	2	2	4

Items table:

item_id	item_brand
1	Samsung
2	Lenovo
3	LG
4	HP

Output:

buyer_id	join_date	orders_in_2019
1	2018-01-01	1
2	2018-02-09	2
3	2018-01-19	0
4	2018-05-21	0

-- Insert data into Users table

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (1, '2017-01-15', 'Sony');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (2, '2018-03-22', 'Apple');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (3, '2019-07-30', 'Samsung');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (4, '2016-11-05', 'Dell');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (5, '2020-02-14', 'HP');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (6, '2019-06-10', 'Lenovo');
```

```
INSERT INTO Users (user_id, join_date, favorite_brand) VALUES (7, '2018-09-25', 'Asus');
```

-- Insert data into Orders table

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (1, '2019-01-12', 2, 1, 3);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (2, '2019-02-15', 1, 2, 4);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (3, '2019-03-20', 4, 1, 5);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (4, '2019-06-25', 3, 3, 2);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (5, '2019-07-30', 5, 4, 6);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (6, '2019-09-10', 2, 6, 7);
```

```
INSERT INTO Orders (order_id, order_date, item_id, buyer_id, seller_id) VALUES (7, '2018-12-05', 1, 5, 2); -- Not in 2019
```

-- Insert data into Items table

```
INSERT INTO Items (item_id, item_brand) VALUES (1, 'Apple');
```

```
INSERT INTO Items (item_id, item_brand) VALUES (2, 'Sony');
```

```
INSERT INTO Items (item_id, item_brand) VALUES (3, 'Samsung');
```

```
INSERT INTO Items (item_id, item_brand) VALUES (4, 'Dell');
```

```
INSERT INTO Items (item_id, item_brand) VALUES (5, 'HP');
```

Question 6. User Orders Count in 2019 | (Medium)

(AMAZON,GOOGLE,UBER)

Write a solution to find for each user,

the join date and the number of orders they made as a buyer in 2019.

Users table:

user_id	join_date	favorite_brand
1	2018-01-01	Lenovo
2	2018-02-09	Samsung
3	2018-01-19	LG
4	2018-05-21	HP

order_id	order_date	item_id	buyer_id	seller_id
1	2019-08-01	4	1	2
3	2019-08-03	3	2	3
6	2019-08-05	2	2	4

user_id	join_date	favorite_brand
1	2018-01-01	Lenovo
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4	2018-08-04	1	4	2
5	2018-08-04	1	3	4
6	2019-08-05	2	2	4

```
SELECT u.user_id as buyer_id,
u.join_date,
count(o.order_id) as orders_in_2019
FROM users u
LEFT JOIN orders o
ON u.user_id = o.buyer_id
AND o.order_date BETWEEN '2019-01-01' AND '2019-12-31'
GROUP BY u.user_id, u.join_date;
```

```

SELECT u.user_id as buyer_id,
u.join_date,
count(o.order_id) as orders_in_2019
FROM users u
LEFT JOIN orders o
ON u.user_id = o.buyer_id
WHERE o.order_date BETWEEN '2019-01-01' AND '2019-12-31'
GROUP BY u.user_id, u.join_date;

```

user_id	join_date	favorite_brand
1	2018-01-01	Lenovo
2	2018-02-09	Samsung
3	2018-01-19	LG
4	2018-05-21	HP

order_id	order_date	item_id	buyer_id	seller_id	join_date	favorite_brand
1	2019-08-01	4	1	2	2018-01-01	Lenovo
3	2019-08-03	3	2	3	2018-02-09	Samsung
6	2019-08-05	2	2	4	2018-02-09	Samsung

1, 1
2, 2

```

SELECT u.user_id as buyer_id,
u.join_date,
count(o.order_id) as orders_in_2019
FROM users u
LEFT JOIN orders o
ON u.user_id = o.buyer_id
AND YEAR(o.order_date) = 2019
-- BETWEEN '2019-01-01' AND '2019-12-31'
GROUP BY u.user_id, u.join_date;
-- Write your PostgreSQL query statement below
SELECT u.user_id as buyer_id,
u.join_date,
count(o.order_id) as orders_in_2019
FROM users u
LEFT JOIN orders o
ON u.user_id = o.buyer_id
AND EXTRACT (YEAR FROM o.order_date) = 2019
-- BETWEEN '2019-01-01' AND '2019-12-31'
GROUP BY u.user_id, u.join_date;

```